

Brain volume changes on longitudinal magnetic resonance imaging in normal older people.



BACKGROUND AND PURPOSE: The purpose of this study is to investigate the longitudinal age-related changes in human brain volume using stereological methods. **METHODS:** Sixty-six older participants (34 men, 32 women, age [mean $\pm$ SD] 78.9 $\pm$ 3.3 years, range 74-87 years) with normal baseline and follow-up examinations underwent 2 MRIs (magnetic resonance imaging) of the brain on average 4.4 years apart. The volumes of the cerebrum (defined as cortex, basal ganglia, thalamus, and white matter), lateral ventricles, and cerebellum were estimated on the 2 MRIs using an unbiased stereological method (Cavalieri principle). **RESULTS:** The annual decrease (mean $\pm$ SD) of the cerebral volume was 2.1% $\pm$ 1.6% ($P < .001$). The average volume of the lateral ventricles on the second MRI was increased by 5.6% $\pm$ 3.6% per year ($P < .001$). The average volume of the cerebellum on the second MRI was decreased by 1.2% $\pm$ 2.2% per year ($P < .001$). Even though the average cerebral volume was significantly different between men and women on initial MRI and second MRI, the percentage change of the age-related cerebral volume decrease in male and female brains between initial MRI and second MRI were identical. **CONCLUSIONS:** The findings showed that there was age-related atrophy of cerebrum and cerebellum and age-related disproportional enlargement of lateral ventricles in normal older men and women.